Berk Yilmaz

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Experience

Software Developer Dec 2024 – Present
Vretta *Toronto, ON*

- Led load testing for provincial assessment platforms across **6** Canadian jurisdictions, running **k6** on **AWS EKS** at up to **6,331 req/s** and **8,041** concurrent students with **zero server errors**, from production traffic analysis through to signed-off client reports.
- Cut a ministry data export from **12.8 hours to 7.2 minutes** (**107**×) by reconstructing the run from **229,530** S3 object timestamps at zero test cost and tracing it to one cross-region S3 write per student.
- Right-sized a production database **two instance sizes** below plan after measuring **25.8%** CPU under **3**× production load, moving the saving to an under-provisioned API tier; recommendation approved by management.
- Closed privilege-escalation gaps in a provincial admin portal as sole developer, re-scoping **35** endpoints shared between school-district and ministry users into role-aware routes across **52 Angular** components, and patching **5** authentication vulnerabilities including a login-key bypass and query-operator injection.
- Deployed a serverless error-alerting pipeline on **AWS Lambda** and **Athena** that joins load-balancer logs to application stack traces by trace ID, rolling it out to **11** tenants across **2** regions under data-residency rules via **GitLab CI**.
- Collapsed production error noise **50**×, turning **3,253** alerts into **65** distinct defects, by fingerprinting failures in **DynamoDB** and opening one tracked GitLab issue per unique bug instead of a duplicate flood.

Full-Stack Web Developer May 2024 – Dec 2024
Expense Trend (Contract) *Calgary, AB*

- Replaced a manual configuration process with a documented REST API in **Node.js** and **OpenAPI**, cutting admin setup time by an estimated **25%**.
- Designed the **MSSQL** schema and persistence layer for synced bank accounts, institutions and credentials, letting users link accounts and view transactions in one flow and cutting login latency roughly **30%**.

Projects

Departure Brief | *TypeScript, React, Fastify, PostgreSQL, Cloudflare Workers, local LLMs* live demo | source

- Built and deployed a preflight briefing web app merging **five** live aviation feeds through a **Cloudflare Worker**, citing the exact source text behind every finding, in **30,200** lines of **TypeScript** with **899** tests in CI.
- Wrote dependency-free parsers for coded weather reports, leaving under **0.3%** of tokens unparsed across **8,000** real reports, and stored raw reports append-only and content-addressed so any past briefing rebuilds exactly.
- Ranked airspace notices with a **locally run AI model** that must quote its source before a claim is accepted, catching **100%** of critical notices with **zero** unverifiable citations at **$0** running cost.
- Caught that the project's AI quality test could not fail: rules settled **22 of 28** cases before the model ran, so a model giving one fixed answer would score **96%**; rebuilt it to grade the model only on what the rules leave open.

Hours to Go | *TypeScript, React, Fastify, PostgreSQL* live demo | source

- Built a pilot logbook web app tracking flight time against **9** Transport Canada licences and ratings, with every requirement encoded from **CAR Standard 421** and cited to its paragraph.
- Added accounts with **scrypt**-hashed passwords and HttpOnly session cookies, instant edits saved through an ordered queue that rolls back on failure, and guest logbooks that merge without duplicates.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Backend: Node.js, Feathers.js, Express, REST APIs, OpenAPI
Frontend: Angular, React
Cloud & Infrastructure: AWS (Lambda, EKS, Athena, DynamoDB, S3, CloudWatch, IAM), Terraform, Kubernetes, Docker, GitLab CI/CD
Data & AI: PostgreSQL, Aurora/MySQL, MSSQL, Redis, k6, LLM structured extraction, evaluation harnesses, OCR
Certifications: Private Pilot Licence (Aeroplane), Transport Canada

Education

Wilfrid Laurier University Waterloo, ON
Honours Bachelor of Science in Computer Science *Sep 2020 – May 2024*